

01 APPOINTMENTS

Postdoctoral Researcher

Sep 2023 – Present

Università della Svizzera italiana (USI), Lugano, Switzerland / Advisor: Prof. Paolo Tonella / Funding: the European Horizon project Sec4AI4Sec and the NRF Korea Postdoctoral Fellowship Program 2024 / Research on testing and assurance for deep-learning and RAG systems.

Postdoctoral Researcher

Mar 2023 – Aug 2023

KAIST, Daejeon, South Korea / Advisor: Prof. Shin Yoo / Postdoctoral research at the COINSE lab.

Visiting Ph.D. Student

Aug 2022 – Sep 2022

Università della Svizzera italiana (USI), Lugano, Switzerland / Advisor: Prof. Paolo Tonella / Funding: Young Researchers' Exchange Programme between South Korea and Switzerland 2022 / Research on mutation-based deep-learning system testing and deep-learning program repair.

Research Intern, Users & Information Lab

Sep 2015 – Dec 2015

KAIST, Daejeon, South Korea / Advisor: Prof. Alice Oh / Designed and deployed EliceQ, a social back-channel application for anonymous classroom questions.

Research Intern, NC Lab

Mar 2015 – Aug 2015

KAIST, Daejeon, South Korea / Advisor: Prof. Junehwa Song / Developed BeUpright and supported a two-week human study on relational-norm interventions for posture change.

02 EDUCATION

Ph.D. in Computer Science

Mar 2017 – Feb 2023

KAIST, Daejeon, South Korea

Integrated M.S./Ph.D. programme. • Advisor: Prof. Shin Yoo.

B.S. in Computer Science

Mar 2012 – Feb 2017

KAIST, Daejeon, South Korea

03 PH.D. SUPERVISION

Masoud Jamshidiyan Tehrani

Feb 2024 – Present

Ph.D. student at USI; co-supervised with Prof. Paolo Tonella

Samuele Pasini

Oct 2023 – Present

Ph.D. student at USI; co-supervised with Prof. Paolo Tonella

04 AWARDS, HONORS & PATENT

2023 **Best Paper Award** 18th International Workshop on Mutation Analysis

2023 **KAIST Ph.D. Dissertation Award** College of Engineering, KAIST

- 2020 **NAVER Ph.D. Fellowship Award** School of Computing, KAIST
- 2021 **Korea Patent No. 1020190104591** *Method for Evaluating Test Fitness of Input Data for Neural Network and Apparatus Thereof*; published August 9, 2021.

05 PUBLICATIONS

Selected papers appear first; Jinhan Kim appears in navy. All are main-track unless noted.

[Google Scholar](#) | 12 Aug 2026 | Citations **1,067** | h-index **12** | i10-index **14**

SELECTED PUBLICATIONS

- [1] **ASE · 2026** **When Knowledge Changes: Metamorphic Testing of RAG Systems with Mutations**
J. Kim, S. Pasini, and P. Tonella • *Proceedings of the 41st IEEE/ACM International Conference on Automated Software Engineering*.
- [2] **ISSTA · 2026** **Testing Retrieval-Augmented Generation Systems with Chunk Coverage**
J. Kim, S. Pasini, and P. Tonella • *Proceedings of the 35th ACM SIGSOFT International Symposium on Software Testing and Analysis*.
- [3] **ICSE · 2026** **Revisiting "Revisiting Neuron Coverage for DNN Testing: A Layer-Wise and Distribution-Aware Criterion": A Critical Review and Implications on DNN Coverage Testing**
J. Kim, N. Humbatova, G. Jahangirova, S. Yoo, and P. Tonella • *Proceedings of the 48th IEEE/ACM International Conference on Software Engineering*.
- [4] **EMSE · 2025** **Evaluating and Improving the Robustness of Security Attack Detectors Generated by LLMs**
S. Pasini, J. Kim, T. Aiello, R. Lozoya, A. Sabetta, and P. Tonella • *Empirical Software Engineering*.
• Journal-first presentation at ICSE 2026
- [5] **TOSEM · 2025** **A Taxonomy of System-Level Attacks on Deep Learning Models in Autonomous Vehicles**
M. J. Tehrani, J. Kim, R. Z. L. Foulefact, A. Marchetto, and P. Tonella • *ACM Transactions on Software Engineering and Methodology*.
• Journal-first presentation at ICSE 2026
- [6] **EMSE · 2025** **An Empirical Study of Fault Localisation Techniques for Deep Neural Networks**
N. Humbatova, J. Kim, G. Jahangirova, S. Yoo, and P. Tonella • *Empirical Software Engineering*, 30(5), p. 124.
• Journal-first presentation at ICST 2026
- [7] **TOSEM · 2023** **Evaluating Surprise Adequacy for Deep Learning System Testing**
J. Kim, R. Feldt, and S. Yoo • *ACM Transactions on Software Engineering and Methodology*, 32(2), pp. 1–29.
- [8] **ICST · 2023** **Repairing DNN Architecture: Are We There Yet?**
J. Kim, N. Humbatova, G. Jahangirova, P. Tonella, and S. Yoo • *Proceedings of the 16th IEEE International Conference on Software Testing, Verification and Validation*, pp. 234–245.
- [9] **TOSEM · 2022** **Predictive Mutation Analysis via Natural Language Channel in Source Code**
J. Kim, J. Jeon, S. Hong, and S. Yoo • *ACM Transactions on Software Engineering and Methodology*, 31(4), pp. 1–27.
• Journal-first presentation at ICSE 2023
- [10] **ICSE · 2019** **Guiding Deep Learning System Testing Using Surprise Adequacy**
J. Kim, R. Feldt, and S. Yoo • *Proceedings of the 41st IEEE/ACM International Conference on Software Engineering*, pp. 1039–1049.

OTHER JOURNAL ARTICLES

- [1] **JSS · 2026** **Cross-Site Scripting Adversarial Attacks Based on Deep Reinforcement Learning: Evaluation and Extension Study**
S. Pasini, G. Maragliano, J. Kim, and P. Tonella • *Journal of Systems and Software*.
- [2] **IST · 2023** **Learning Test-Mutant Relationship for Accurate Fault Localisation**
J. Kim, G. An, R. Feldt, and S. Yoo • *Information and Software Technology*, 162, p. 107272.

OTHER CONFERENCES & WORKSHOPS

- [1] **TRUST · 2026** **Dynamic Deception: When Pedestrians Team Up to Fool Autonomous Cars**
M. Tehrani, M. Gabriel, **J. Kim**, and P. Tonella • *Proceedings of the 1st International Workshop on Trustworthy and Responsible aUtonomous SysTems*.
- [2] **FSE Demo · 2025** **PCLA: A Framework for Testing Autonomous Agents in the CARLA Simulator**
M. Tehrani, **J. Kim**, and P. Tonella • *Proceedings of the 33rd ACM International Conference on the Foundations of Software Engineering*, pp. 1040–1044.
- [3] **Mutation · 2023** **The Inversive Relationship Between Bugs and Patches: An Empirical Study**
J. Kim, J. Park, and S. Yoo • *Proceedings of the 18th International Workshop on Mutation Analysis*, pp. 314–323.
- [4] **ICST Industry · 2022** **Repairing Fragile GUI Test Cases Using Word and Layout Embedding**
J. Yoon, S. Chung, K. Shin, **J. Kim**, S. Hong, and S. Yoo • *Proceedings of the 15th IEEE International Conference on Software Testing, Verification and Validation*.
- [5] **ISSRE · 2021** **Ahead of Time Mutation Based Fault Localisation Using Statistical Inference**
J. Kim, G. An, R. Feldt, and S. Yoo • *Proceedings of the 32nd International Symposium on Software Reliability Engineering*, pp. 253–263.
- [6] **FSE Industry · 2020** **Reducing DNN Labelling Cost Using Surprise Adequacy: An Industrial Case Study for Autonomous Driving**
J. Kim, J. Ju, R. Feldt, and S. Yoo • *Proceedings of the 28th ACM Joint Meeting on European Software Engineering Conference and Symposium on the Foundations of Software Engineering*, pp. 1466–1476.
- [7] **SSBSE · 2018** **Learning Without Peeking: Secure Multi-Party Computation Genetic Programming**
J. Kim, M. G. Epitropakis, and S. Yoo • *Proceedings of the 10th International Symposium on Search Based Software Engineering*, pp. 246–261.
- [8] **L@S · 2018** **Elicast: Embedding Interactive Exercises in Instructional Programming Screencasts**
J. Park, Y. H. Park, **J. Kim**, J. Cha, S. Kim, and A. Oh • *Proceedings of the Fifth Annual ACM Conference on Learning at Scale*, pp. 1–10.
- [9] **GI · 2018** **Comparing Line and AST Granularity Level for Program Repair Using PyGGI**
G. An, **J. Kim**, and S. Yoo • *Proceedings of the 4th Genetic Improvement Workshop*, pp. 19–26.
- [10] **KCSE · 2017** **PyGGI: Python General framework for Genetic Improvement**
G. An, **J. Kim**, S. Lee, and S. Yoo • *Proceedings of Korea Software Congress*, pp. 536–538.
- [11] **SSBSE Short Papers · 2017** **GPGPGPU: Evaluation of Parallelisation of Genetic Programming Using GPGPU**
J. Kim, J. Kim, and S. Yoo • *Proceedings of the 9th International Symposium on Search Based Software Engineering*, pp. 137–142.
- [12] **CHI · 2016** **BeUpright: Posture Correction Using Relational Norm Intervention**
J. Shin, B. Kang, T. Park, J. Huh, **J. Kim**, and J. Song • *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems*, pp. 6040–6052.
- [13] **SenSys Demo · 2015** **Demo: Posture Correction Using Smartphone-Based Relational Intervention Model**
J. Shin, B. Kang, **J. Kim**, J. Huh, J. Song, and T. Park • *Proceedings of the 13th ACM Conference on Embedded Networked Sensor Systems*, pp. 495–496.

PREPRINTS & MANUSCRIPTS

- [1] **arXiv · 2026** **Detecting Trojaned DNNs via Spectral Regression Analysis**
S. Pasini, **J. Kim**, and P. Tonella • *arXiv preprint*.
- [2] **arXiv · 2025** **Real Faults in Deep Learning Fault Benchmarks: How Real Are They?**
G. Jahangirova, N. Humbatova, **J. Kim**, S. Yoo, and P. Tonella • *arXiv preprint*.
• Under revision at *Empirical Software Engineering*
- [3] **arXiv · 2025** **Mutation-Guided Test Generation for DNNs Using Diffusion**
S. Kim, **J. Kim**, V. Riccio, S. Yoo, and P. Tonella • *arXiv preprint*.
- [4] **arXiv · 2025** **MuFF: Stable and Sensitive Post-training Mutation Testing for Deep Learning**
J. Kim, N. Humbatova, G. Jahangirova, S. Yoo, and P. Tonella • *arXiv preprint*.
- [5] **arXiv · 2025** **Fault Localisation and Repair for DL Systems: An Empirical Study with LLMs**
J. Kim, N. Humbatova, G. Jahangirova, S. Yoo, and P. Tonella • *arXiv preprint*.
• Under revision at *ACM Transactions on Software Engineering and Methodology*

06 ACADEMIC SERVICE

Organising Committees

- 2027** ICST, Posters Track Co-chair
- 2026** ICST, Testing Tools and Data Showcase Track Co-chair; SSBSE, Publicity Co-chair; DeepTest, Workshop Co-organiser; SBFT, Workshop Co-organiser
- 2025** DeepTest, Workshop Co-organiser

Program Committees

- 2027** ICSE, Research Track
- 2026** ASE, Research Track; ESEM, Research Track; ICST, Education Track; Mutation; RASE
- 2025** ISSTA, Research Track; ASE, NIER Track and Student Research Competition; SSBSE, Challenge Track; SBFT; Mutation; DeMeSSAI
- 2024** SCAM, Research Track; ICSE, Demonstrations Track; ASE, Demonstrations and NIER Tracks; DeepTest; Mutation
- 2023** Mutation; ASE, NIER Track; ICSME, Artifact Evaluation Track and ROSE Festival
- 2022** Mutation; ICST, Poster Track; ICSME, Registered Reports Track, Artifact Evaluation Track, and ROSE Festival
- 2021** Mutation; ICSME, Artifact Evaluation Track
- 2020** Mutation

Journal Boards & Reviewing

- 2026** Reviewer: Empirical Software Engineering
- 2025** TOSEM Board of Distinguished Reviewers; reviewer for TSE, TOSEM, EMSE, STVR, and IEEE Computer
- 2024** TOSEM Board of Distinguished Reviewers; reviewer for TSE, TOSEM, EMSE, JSEP, and JSS
- 2023** Reviewer for TSE, TOSEM, EMSE, and JSEP
- 2022** Reviewer for TOSEM and JSEP
- 2021** Reviewer for TOSEM and STVR
- 2020** Reviewer for IST and JSS

07 INVITED TALKS

- 2026** **Mutation Workshop, ICST 2026, Daejeon, South Korea** / Panellist.
- 2026** **Software Institute Seminar, USI, Lugano, Switzerland** / *Testing RAG Systems: Coverage-Guided Adequacy and Metamorphic Oracles.*
- 2026** **CHOOSE Forum, Bern, Switzerland** / *Breaking DNNs: Mutation Testing for Deep Neural Networks.*
- 2024** **Software Institute Seminar, USI, Lugano, Switzerland** / *When Simple is Better than Complex: Coverage and Mutation for DL Testing.*
- 2023** **Korea Conference on Software Engineering, Pyeongchang, South Korea** / Invited paper presentation: *Predictive Mutation Analysis via Natural Language Channel in Source Code.*
- 2022** **Korea Conference on Software Engineering, Pyeongchang, South Korea** / Invited paper presentation: *Ahead of Time Mutation Based Fault Localisation using Statistical Inference.*
- 2019** **Korea Software Congress, Pyeongchang, South Korea** / Invited paper presentation: *Guiding Deep Learning System Testing Using Surprise Adequacy.*

08 TEACHING

KAIST CS101: Introduction to Programming – Teaching Assistant, Fall 2016, Spring 2020, Fall 2020, Spring 2021.

KAIST CS453: Automated Software Testing – Teaching Assistant, Spring 2018, Spring 2019.

KAIST CS454: AI Based Software Engineering – Teaching Assistant, Fall 2021.

KAIST CS489: Computer Ethics and Social Issues – Teaching Assistant, Fall 2019.

09 REFERENCES

<u>Prof. Shin Yoo</u>	Full Professor, KAIST • Daejeon, South Korea	shin.yoo@kaist.ac.kr
<u>Prof. Paolo Tonella</u>	Full Professor, USI • Lugano, Switzerland	paolo.tonella@usi.ch
<u>Prof. Robert Feldt</u>	Full Professor, Chalmers • Gothenburg, Sweden	robert.feldt@chalmers.se